

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-1116

Clover 345 kV
November 16, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1116. The Interconnection Customer proposes a 880.4 MW natural gas combined cycle facility interconnecting at the Clover 345 kV substation in the Virginia area. The request was received on November 27, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 452 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$53,149,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
HUTCHINS - CLOVER 138kV	CLOVER - BEAR 345kV	162.4	464	14.6
KEOKUK - KENBRIDGE 138kV Line	BEAR - CLOVER SUBSTATION 345kV	139.8	435	7.5
HUTCHINS - BAUXITE 138kV	BEAR - CLOVER 345kV	139.5	491	10.0
KENBRIDGE - KLAMATH FALLS 138kV	BEAR SUBSTATION - CLOVER 345kV L	139.0	441	7.5
KLAMATH FALLS - BAUXIE 138kV line	BEAR - CLOVER 345kV L	138.2	448	7.5
CLOVER 345/138kV Auto	CLOVER - BEAR 345kV	118.2	495	14.6
S CREEK - AIBONITO PDS 138kV	BEAR - CLOVER 345kV	105.8	549	6.3
LEAVITT - BEAR 345kV Line 2	BEAR - LEAVITT 345kV #1	105.3	2,331	22.5
CHANCELLOR ROW - LEAVITT SUBSTATION 345kV	BIRCHWOOD - LEAVITT 345kV	104.4	1,714	17.5
STATE SCHOOL - TRINIDAD 138kV	CLOVER - BEAR 345kV	104.0	419	5.9
MECKLENBURG POWER STATION - KERR DAM 345kV	MECKLENBURG POWER STA - BEAR 345kV	100.7	1,696	11.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HUTCHINS - CLOVER 138kV	Rebuild line with higher-capacity conductor	\$22,565,000	55% (\$12,470,000)
KEOKUK - KENBRIDGE 138kV Line	Rebuild line with higher-capacity conductor	\$7,444,000	56% (\$4,190,000)
HUTCHINS - BAUXITE 138kV	Rebuild line with higher-capacity conductor	\$6,872,000	55% (\$3,742,000)
KENBRIDGE - KLAMATH FALLS 138kV	Rebuild line with higher-capacity conductor	\$7,516,000	56% (\$4,231,000)
KLAMATH FALLS - BAUXIE 138kV line	Rebuild line with higher-capacity conductor	\$6,799,000	56% (\$3,827,000)
CLOVER 345 kV/138 kV Autotransformer	Replace transformer with higher-rated unit	\$19,496,000	42% (\$8,290,000)
S CREEK - AIBONITO PDS 138kV	Reconductor line	\$2,593,000	19% (\$493,000)
BEAR - LEAVITT 345kV Ckt 2	Reconductor line	\$64,041,000	16% (\$10,511,000)
CHANCELLOR ROW - LEAVITT SUBSTATION 345kV	Reconductor line	\$9,133,000	15% (\$1,374,000)
TRINIDAD - STATE SCHOOL 138kV	Reconductor line	\$2,263,000	46% (\$1,038,000)
MECKLENBURG POWER STATION - KERR DAM 345kV	Reconductor line	\$18,151,000	16% (\$2,983,000)
	Total allocated to project		\$53,149,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.